**This assignment will not be graded and is only for practice.**

## Level 1

1) Consider the following expression: $v_1 = (1, 0), v_2 = (0, 0), v_3 = (0, 1), v_5 = 1, v_6 = 5, v_7 = (1, 0, 1), v_8 = (0, 0, 1), v_9 = (0, 0, 0), v_{10} = (0, 1, 0)$. Choose the set of correct options. ***1 point***

- [ ] $v_1, v_7$ represent vectors in $\mathrm{R}^2$.
- [ ] $v_2, v_5$ represent vectors in $\mathrm{R}^1$.
- [ ] $v_2, v_5, v_9$ represent vectors in $\mathrm{R}^3$.
- [ ] $v_1, v_2, v_3$ represent vectors in $\mathrm{R}^2$.
- [ ] $v_5, v_6$ represent vectors in $\mathrm{R}^1$.
- [ ] $v_1, v_6$ represent vectors in $\mathrm{R}^1$.
- [ ] $v_8, v_9, v_{10}$ represent vectors in $\mathrm{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$v_1, v_2, v_3$ represent vectors in $\mathrm{R}^2$.
$v_5, v_6$ represent vectors in $\mathrm{R}^1$.
$v_8, v_9, v_{10}$ represent vectors in $\mathrm{R}^3$.

2) Let $v, v_1, v_2$ represent vectors in a vector space $V$ over $\mathrm{R}$. Which of the following expressions make sense and represent vectors in $V$? ***1 point***

- [ ] $cv \in V$, where $c \in \mathrm{R}$.
- [ ] $\frac{v}{2}$.
- [ ] $v^2$.
- [ ] $v_1 - v_2$.
- [ ] $v_1^2 - v_2^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$cv \in V$, where $c \in \mathrm{R}$.
$\frac{v}{2}$.
$v_1 - v_2$.

3) Consider the set $V = \{(z - x, -y) \mid x, y, z \in \mathrm{R}\} \subseteq \mathrm{R}^2$ with the usual addition and scalar multiplication as in $\mathrm{R}^2$ and associated statements given below. Choose the correct options. ***1 point***

- [ ] $V$ is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
- [ ] $V$ is closed under scalar multiplication i.e $(c \in \mathrm{R}, v \in V \implies cv \in V)$.
- [ ] $V$ has a zero element with respect to addition. i.e., there exists some element $0$ such that $v + 0 = v$, for all $v \in V$.
- [ ] $V$ is not a vector space.
- [ ] $(a + b)v = av + bv$ where $a, b \in \mathrm{R}$ and $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V$ is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
$V$ is closed under scalar multiplication i.e $(c \in \mathrm{R}, v \in V \implies cv \in V)$.
$V$ has a zero element with respect to addition. i.e., there exists some element $0$ such that $v + 0 = v$, for all $v \in V$.
$(a + b)v = av + bv$ where $a, b \in \mathrm{R}$ and $v \in V$.

4) Consider a set $V = \{(1, x) \mid x \in \mathrm{R}\} \subseteq \mathrm{R}^2$ with the usual addition and scalar multiplication as in $\mathrm{R}^2$ and associated statements given below. Choose the correct options. ***1 point***

- [ ] $V$ is closed under addition.
- [ ] $V$ has no zero element with respect to addition. i.e., there does not exists any element $0$ such that $v + 0 = v$, for all $v \in V$.
- [ ] $(a + b)v = av + bv$ where $a, b \in \mathrm{R}$ and $v \in V$.

- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V$ has no zero element with respect to addition. i.e., there does not exists any element $0$ such that $v + 0 = v$, for all $v \in V$.
$(a + b)v = av + bv$ where $a, b \in \mathrm{R}$ and $v \in V$.

5) Which of the following sets with the given addition and scalar multiplication (scalars are real numbers in every case) form vector spaces? ***1 point***

$$V_1 = \{(x, y, z) \mid x, y, z \in \mathrm{R}\}$$
$$\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1);$$
$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$$
$$\textit{Scalar multiplication: } c(x, y, z) = (x, y, cz);\ (x, y, z) \in V_1,\ c \in \mathrm{R}$$

$$V_2 = \{(x, y, z) \mid x, y, z \in \mathrm{R}\}$$
$$\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1);$$
$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$$
$$\textit{Scalar multiplication: } c(x, y, z) = (x, cy, z);\ (x, y, z) \in V_2,\ c \in \mathrm{R}$$

$$V_3 = \{(x, y, z) \mid x, y, z \in \mathrm{R}\}$$
$$\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0);$$
$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$$
$$\textit{Scalar multiplication: } c(x, y, z) = (cx, y, z);\ (x, y, z) \in V_3,\ c \in \mathrm{R}$$

Choose the correct options.

- [ ] $V_3$ is not vector space.
- [ ] $V_2$ is vector spaces.
- [ ] All are vector spaces.
- [ ] $V_1$ is not vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V_3$ is not vector space.
$V_1$ is not vector space.

6) If $(a, b, c)$ is the inverse element of $(1, 2, 5) \in \mathrm{R}^3$ of the vector space $\mathrm{R}^3$ , then find the value of $a - b + c$.

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -4

***1 point***

**Level 2**

7) Consider the following sets: ***1 point***

$V_1 = \{A \mid A \in M_{2\times 2}(\mathrm{R})$ and $A$ is a symmetric matrix, i.e., $A = A^T\}$
$V_2 = \{A \mid A \in M_{2\times 2}(\mathrm{R})$ and $A$ is a scalar matrix$\}$
$V_3 = \{A \mid A \in M_{2\times 2}(\mathrm{R})$ and $A$ is a diagonal matrix$\}$
$V_4 = \{A \mid A \in M_{2\times 2}(\mathrm{R})$ and $A$ is a upper triangular matrix$\}$
$V_5 = \{A \mid A \in M_{2\times 2}(\mathrm{R})$ and $A$ is a lower triangular matrix$\}$

Choose the set of correct options.

- [ ] Only $V_1$ is a subspace of $M_{2\times 2}(\mathrm{R})$.
- [ ] Only $V_4$ is a subspace of $M_{2\times 2}(\mathrm{R})$.
- [ ] Both $V_2$ and $V_3$ are subspaces of $M_{2\times 2}(\mathrm{R})$
- [ ] All are subspaces of $M_{2\times 2}(\mathrm{R})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both $V_2$ and $V_3$ are subspaces of $M_{2\times 2}(\mathrm{R})$
All are subspaces of $M_{2\times 2}(\mathrm{R})$

8) Choose the set of correct options. ***1 point***

- [ ] A vector in $\mathrm{R}^3$ can be thought of as matrix of order $1 \times 3$ or $3 \times 1$.
- [ ] Intersection of any two subspaces of a vector space $V$ is not a subspace.
- [ ] Intersection of any two subsets of a vector space $V$ is a subspace.
- [ ] The empty subset of a vector space $V$ is a subspace of $V$.
- [ ] $V = \{(x, 0, 0) \mid x \in \mathrm{R}\}$ is a subspace of $\mathrm{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A vector in $\mathrm{R}^3$ can be thought of as matrix of order $1 \times 3$ or $3 \times 1$.
$V = \{(x, 0, 0) \mid x \in \mathrm{R}\}$ is a subspace of $\mathrm{R}^3$.

9) Consider a set $V = \{(x, y) \mid x, y \in \mathrm{R}\} \subseteq \mathrm{R}^2$ with the usual addition as in $\mathrm{R}^2$ and scalar multiplication is defined as ***1 point***

$$c(x, y) = \{ \begin{matrix} (0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}) & c = 0 \end{matrix} \qquad (x, y) \in V,\ c \in \mathrm{R}$$

Consider the statements given below.

**P:** $V$ is closed under addition.
**Q:** $V$ has zero element with respect to addition. i.e., there exists some element $0$ such that $v + 0 = v$, for all $v \in V$.
**R:** $1.v = v$ where $1 \in \mathrm{R}$ and $v \in V$.
**S:** $a(v_1 + v_2) = av_1 + av_2$ where $v_1, v_2 \in V$ and $a \in \mathrm{R}$.
**T:** $(a + b)v = av + bv$ where $a, b \in \mathrm{R}$ and $v \in V$.

Choose the set of correct options.

- [ ] Only P is true.
- [ ] Only Q is true.
- [ ] Both P and Q are true.
- [ ] P,Q and S are true.
- [ ] Both R and T are not true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both P and Q are true.
P,Q and S are true.
Both R and T are not true.

10) Consider the following statements

**Statement P:** $W = \{A \mid A \in M_{2\times 2}(\mathrm{R}) \text{ and } A^2 = A\}$, a subset of $M_{2\times 2}(\mathrm{R})$ with usual addition and scalar multiplication as in $M_{2\times 2}(\mathrm{R})$, is a subspace of $M_{2\times 2}(\mathrm{R})$.
**Statement Q:** $W = \{A \mid A \in M_{2\times 2}(\mathrm{R}) \text{ and } det(A) = 0\}$, a subset of $M_{2\times 2}(\mathrm{R})$ with usual addition and scalar multiplication as in $M_{2\times 2}(\mathrm{R})$, is a subspace of $M_{2\times 2}(\mathrm{R})$.
**Statement R:** Consider a system of linear equations $Ax^T = b$, where $A$ is invertible square matrix of order 2, $x = (x_1, x_2)$ and $b = 0$. Let $W = \{x = (x_1, x_2) \mid x \in \mathrm{R}^2 \text{ and } Ax^T = b\}$ is a subset of $\mathrm{R}^2$ with usual addition and scalar multiplication as in $\mathrm{R}^2$, is a subspace of $\mathrm{R}^2$.
**Statement S:** Consider a system of linear equations $Ax^T = 0$, where $A$ is invertible square matrix of order 2 and $x = (x_1, x_2)$. Let $W = \{x = (x_1, x_2) \mid x \in \mathrm{R}^2 \text{ and } Ax^T = 0\}$ is a subset of $\mathrm{R}^2$ with usual addition and scalar multiplication as in $\mathrm{R}^2$, is a subspace of $\mathrm{R}^2$.

Find the number of correct statements

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No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

***1 point***

Your score is: 0/10